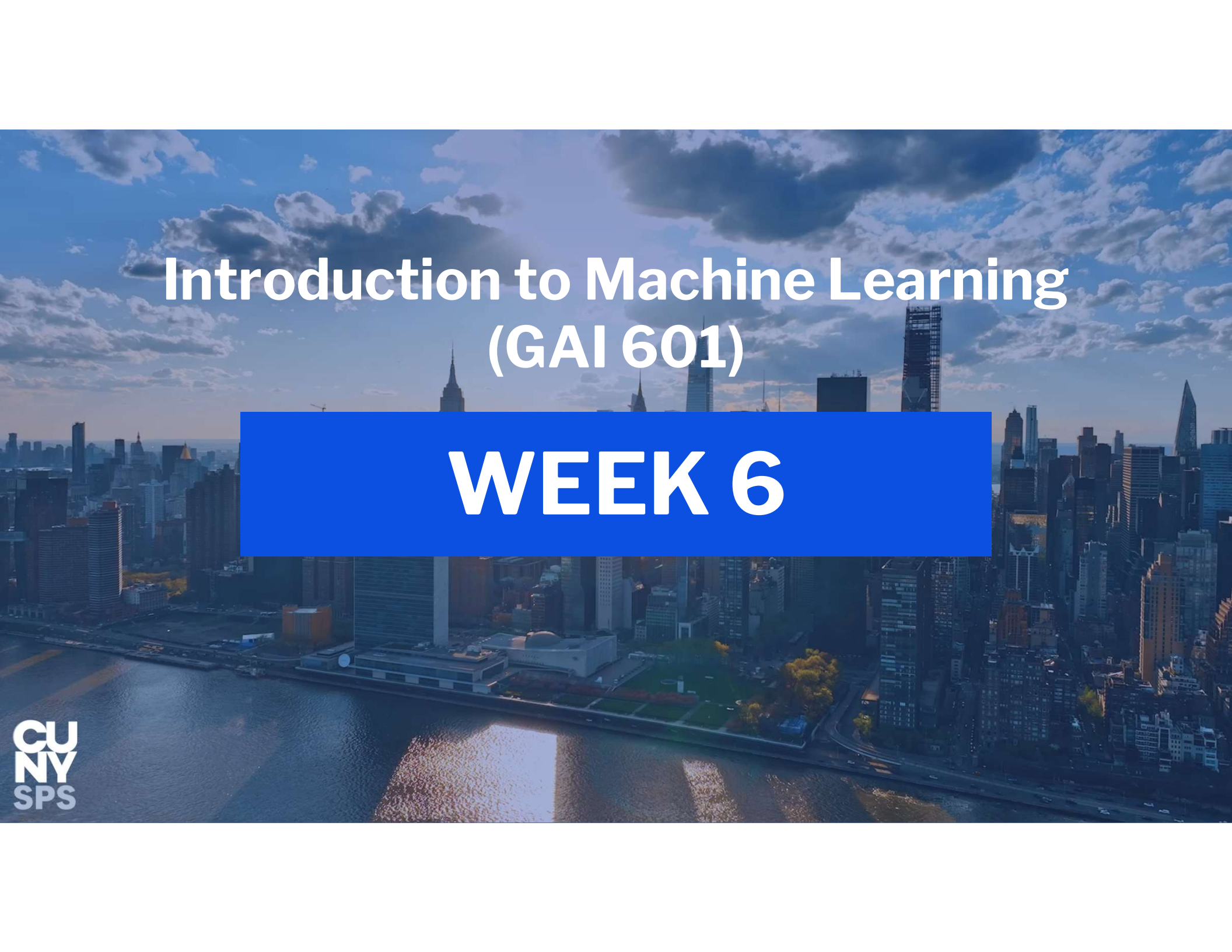


An aerial photograph of the New York City skyline, featuring numerous skyscrapers and the Hudson River in the foreground. A large, semi-transparent blue rectangular box is centered over the image, containing white text. The sky is filled with scattered white clouds.

Introduction to Machine Learning (GAI 601)

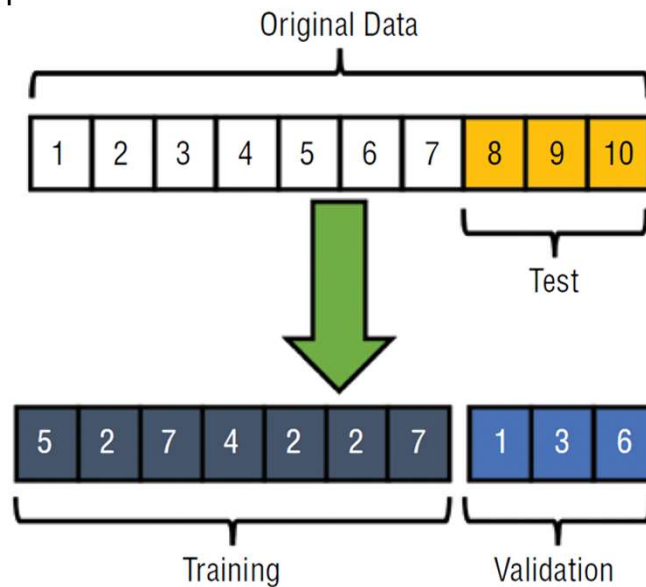
WEEK 6

Bootstrap Sampling



Bootstrap Sampling

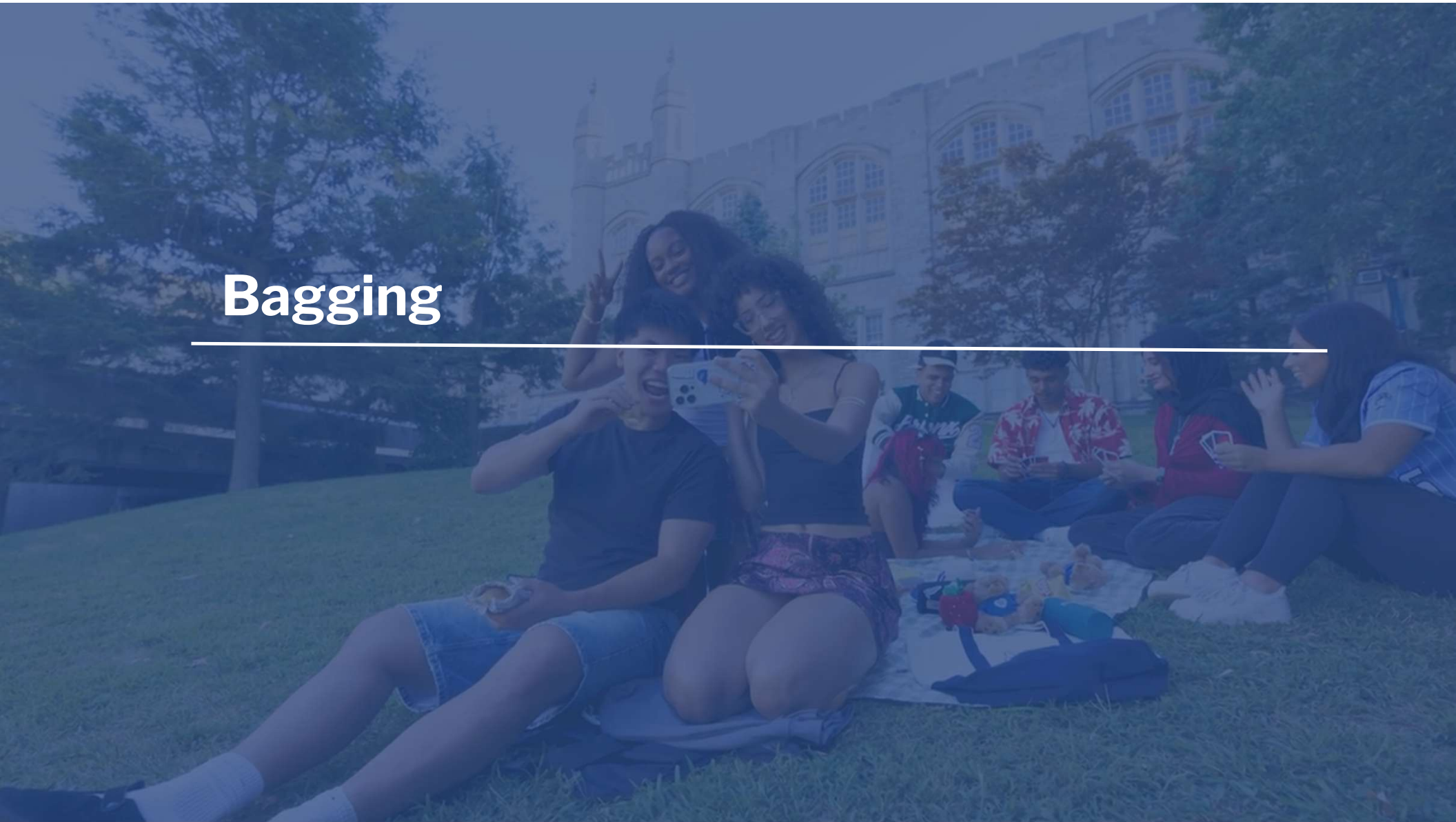
- Create a new training dataset from the original data
- Randomly sample data, **with replacement**
- Every instance has equal chance of being picked
- Some instances may be picked multiple times; others may not be chosen



62.3% of datapoints are unique

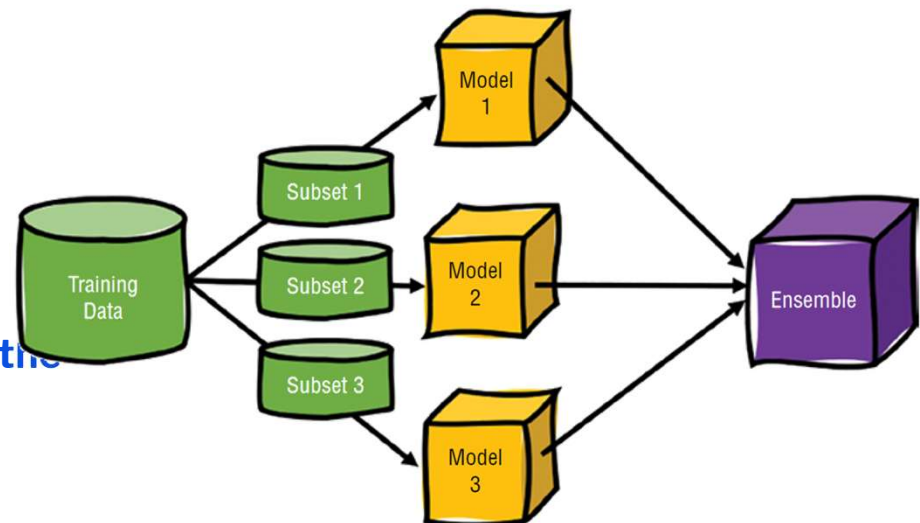
- The probability of picking one row out of n is $\frac{1}{n}$
- Therefore the probability of not picking it is $1 - \frac{1}{n}$
- After n trials the probability of not picking it is $n \left(1 - \frac{1}{n}\right)$
- As n approaches infinity this becomes $e^{-1} = 0.368$
- Hence, approximately 63.2% of datapoints are uniquely selected in a bootstrap

Bagging



Bagging: Bootstrap Aggregation

- Ensemble Learning Method
Combining the results of multiple learners to get a generalized result
- Multiple learners of the **same algorithm** are trained with bootstrapped data
- Works by iteratively generating learners and adding them to the ensemble
- Iteration stops when a predefined number of learners have been added
- In Bagging the result is obtained by **averaging the responses** of the N learners (or majority vote)
- Example: Random Forest

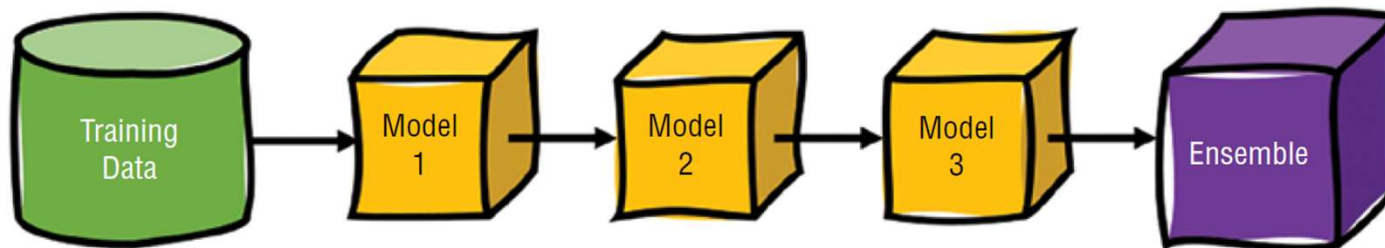


Boosting



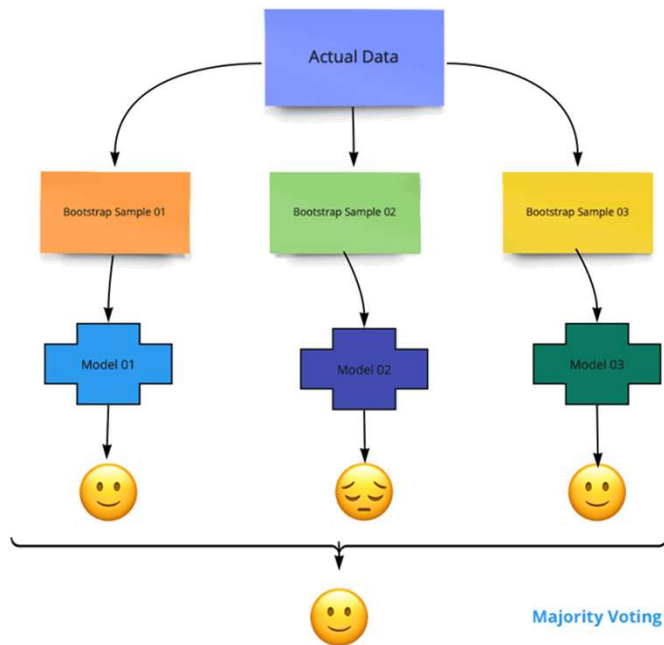
Boosting

- Boosting works by iteratively creating models and adding them to the ensemble
- Iteration stops when a predefined number of models have been added
- Each new model added to the ensemble is biased to pay more attention to instances that previous models misclassified (weighted dataset). A weak learner is trained based on the results of the previous learner (model)
- Two main types:
 - **Adaptive Boosting:** Re-weight datapoints based on performance of last weak learner. Focuses on points where previous learner had trouble. Example: AdaBoost.
 - **Gradient Boosting:** Train new learner on residuals of overall model. Constitutes gradient boosting because approximating the residual and adding to the previous result is essentially a form of gradient descent. Example: XGBoost.



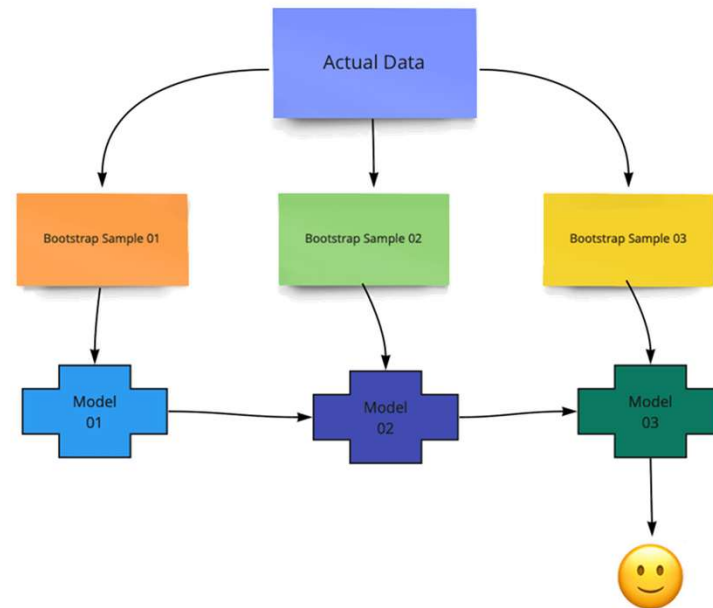
Boosting vs Bagging

Bagging Ensemble Method



VS

Boosting Ensemble Method



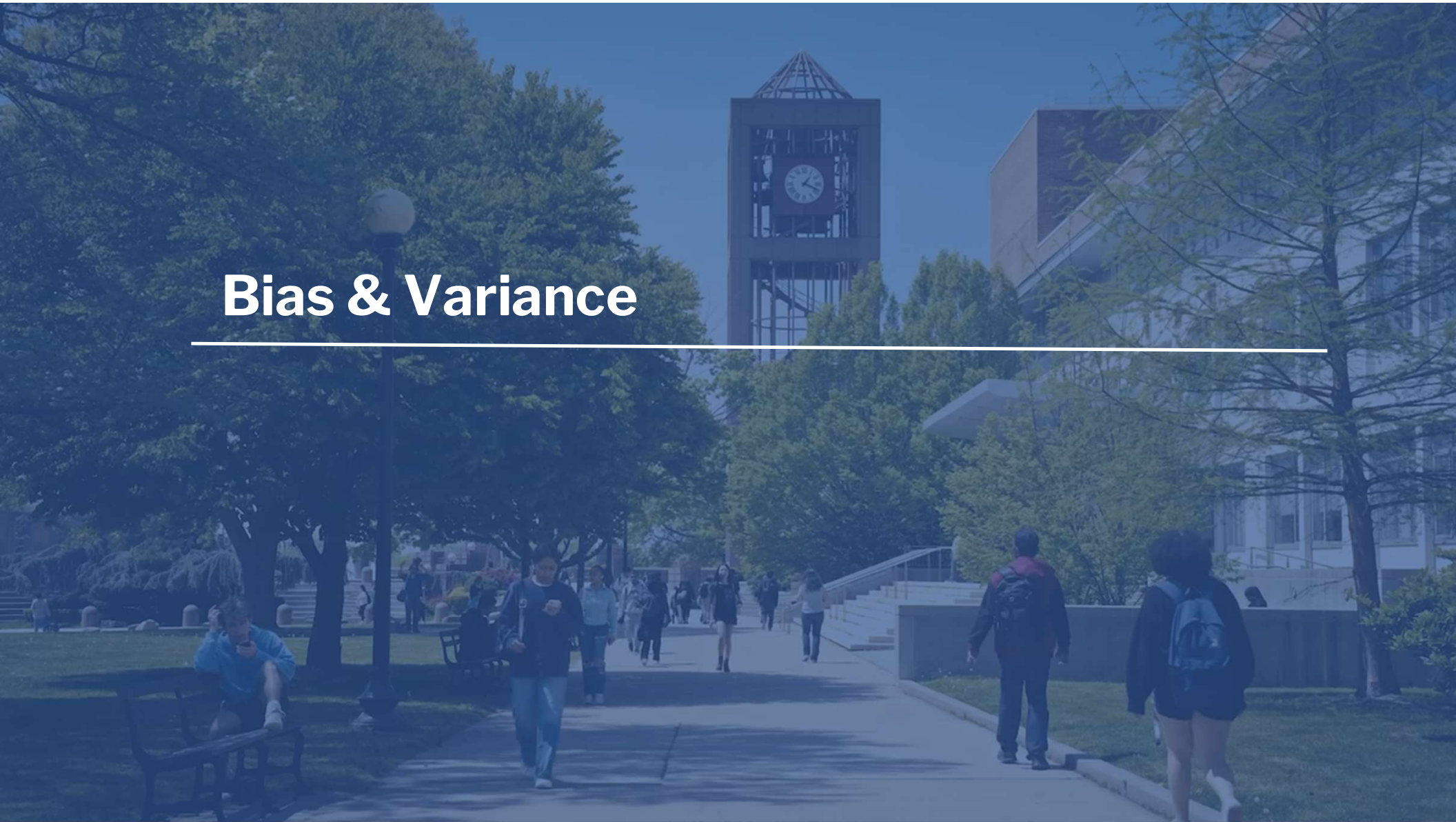
Boosting vs Bagging

- Both are Ensemble Learning methods
- **Bagging**
 - Bagging helps to decrease the model's **variance**
 - Bagging helps in reducing the overfitting as we are averaging all the models outputs using majority voting
- **Boosting**
 - Boosting helps to decrease the model's **bias**
 - Each model tries to reduce the errors of the previous model in the chain
 - Boosting can use multiple loss functions to reduce the error in each sequential models
 - Boosting method may overfit as the models build sequentially (all models try to reduce the training error)
- **Regression:**
 - Bagging: For regression models the predicted value won't be optimized if any one of the models is deviating more as the output value will be average of all the models.

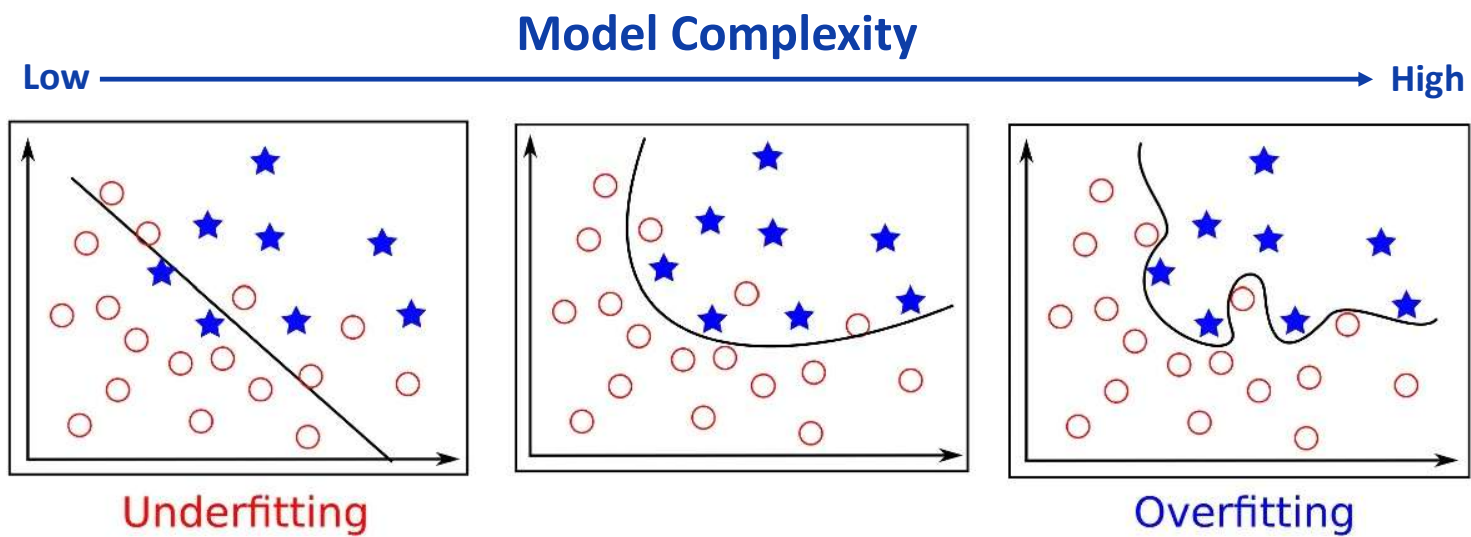
Boosting vs Bagging

- **Different data partition**
 - Bagging uses random samples of the training data for all models independently
 - Boosting puts higher importance on misclassified data of the upcoming models.
- **Independent vs chain models** | (Independent vs. sequence)
 - Bagging creates independent models that are aggregated together.
 - Boosting updates the existing model with the new ones in a sequence.
- **Variance vs. bias**
 - Bagging aims to reduce the variance, but boosting tries to reduce the bias. Therefore, bagging can help to decrease overfitting, and boosting can reduce underfitting.
- **Function** (Weighted vs. non-weighted)
 - For bagging the function to predict the outcome uses equally weighted average/voting aggregations
 - Boosting uses weighted majority vote or weighted average functions with more weight to those with better performance on training data.

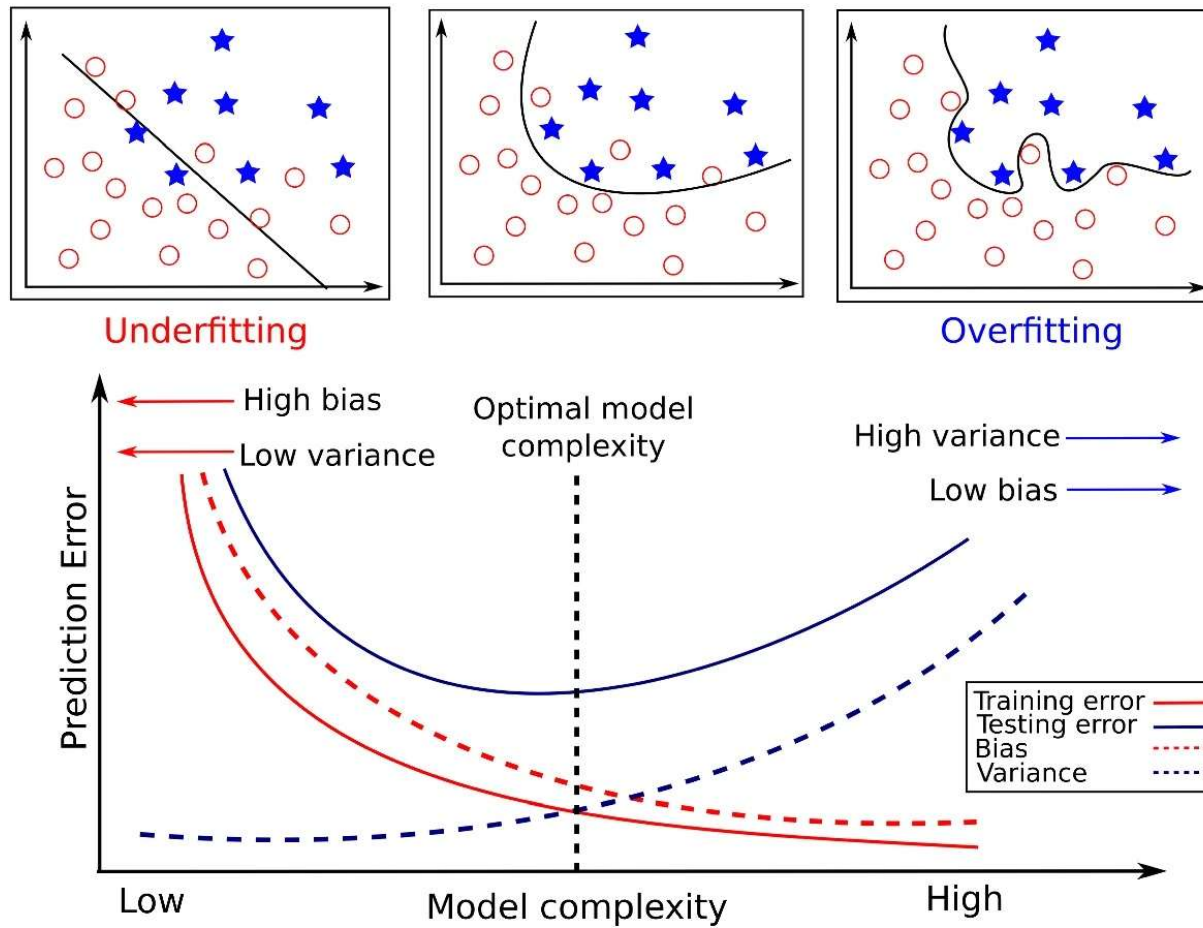
Bias & Variance



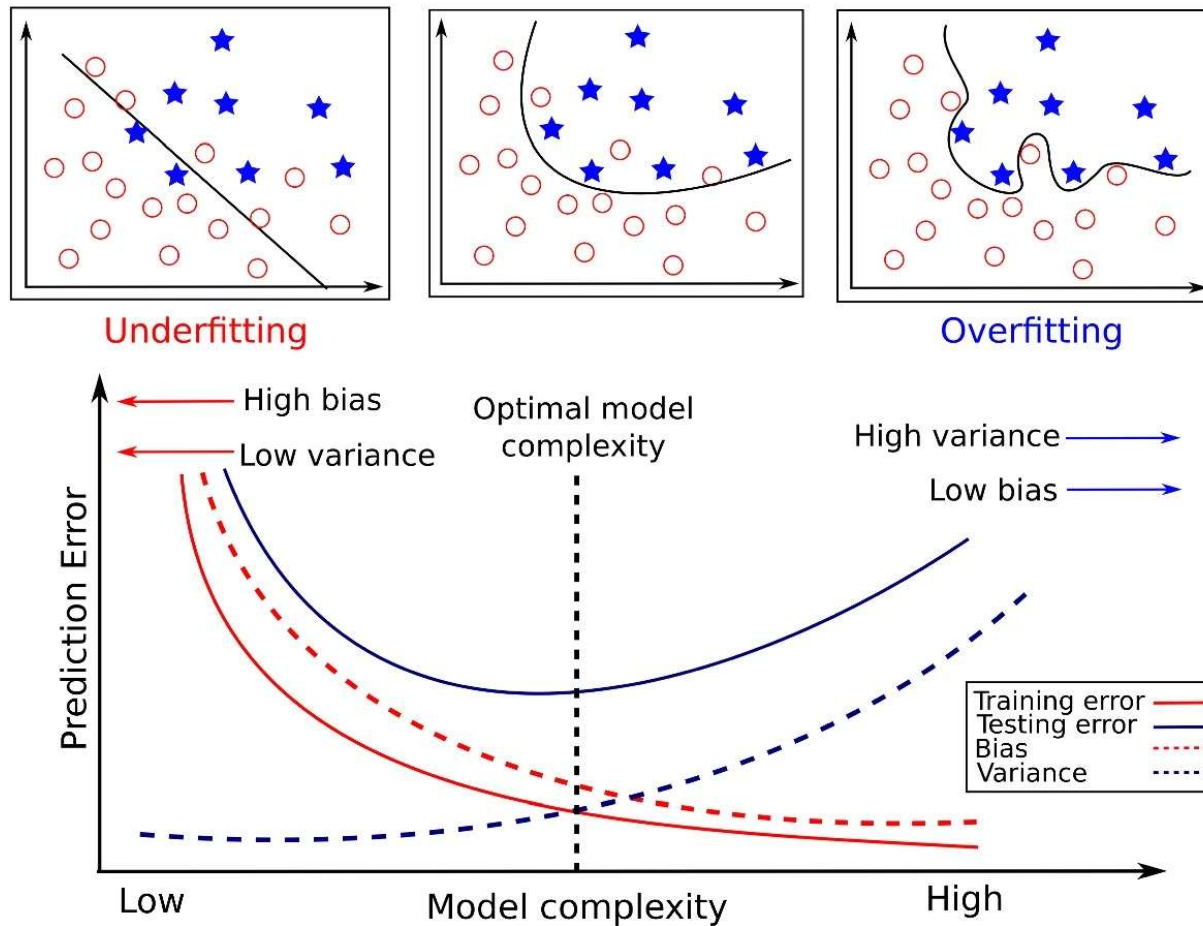
Model Complexity



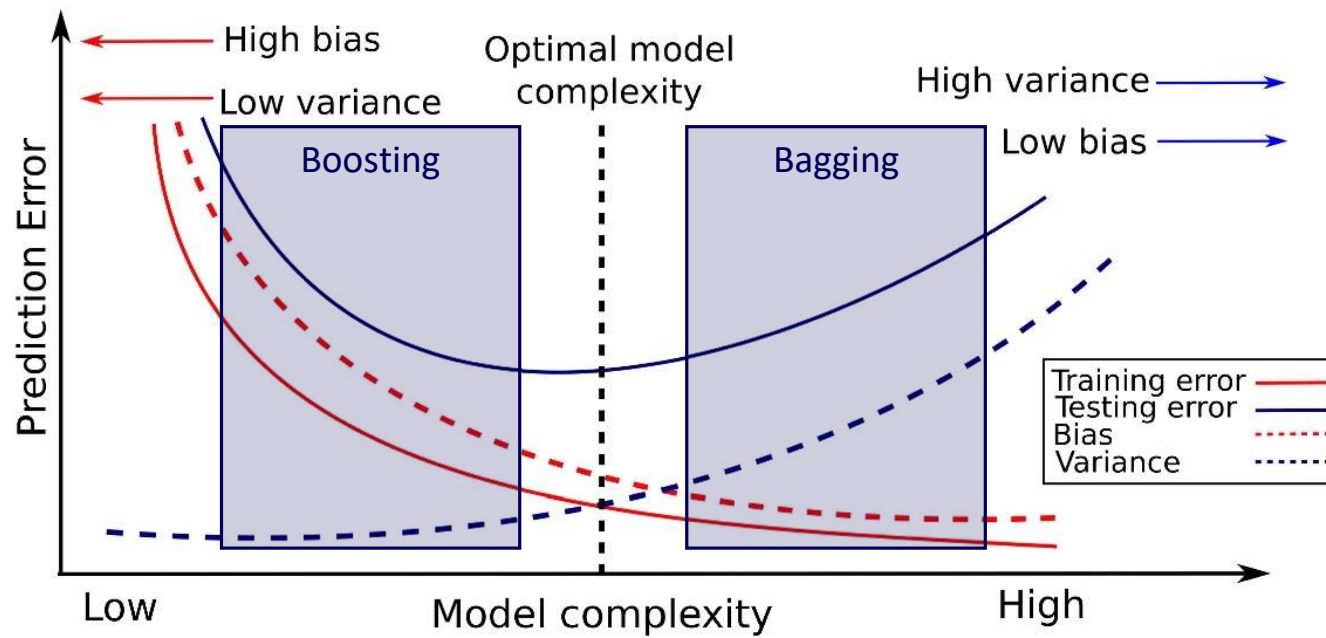
Bias & Variance



Bias & Variance



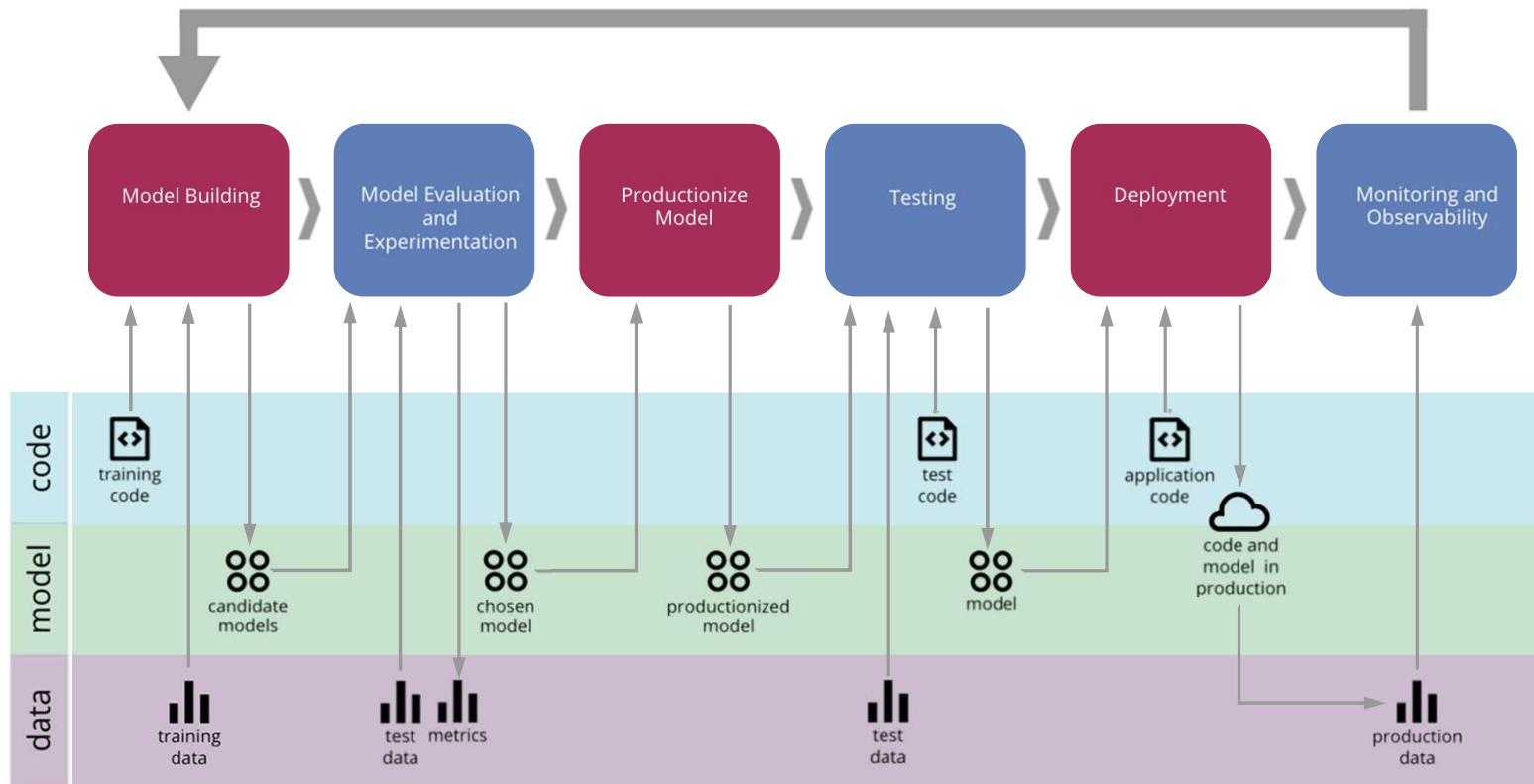
Bias & Variance



A blue-tinted photograph of a university campus. In the background, a prominent clock tower with a glass-enclosed upper section and a clock face is visible. The tower is surrounded by lush green trees. In the foreground, a wide, paved pedestrian walkway leads towards the tower. Several students are walking along the path; some are carrying backpacks. To the left, a person is sitting on a wooden bench. The overall scene is bright and clear, suggesting a sunny day.

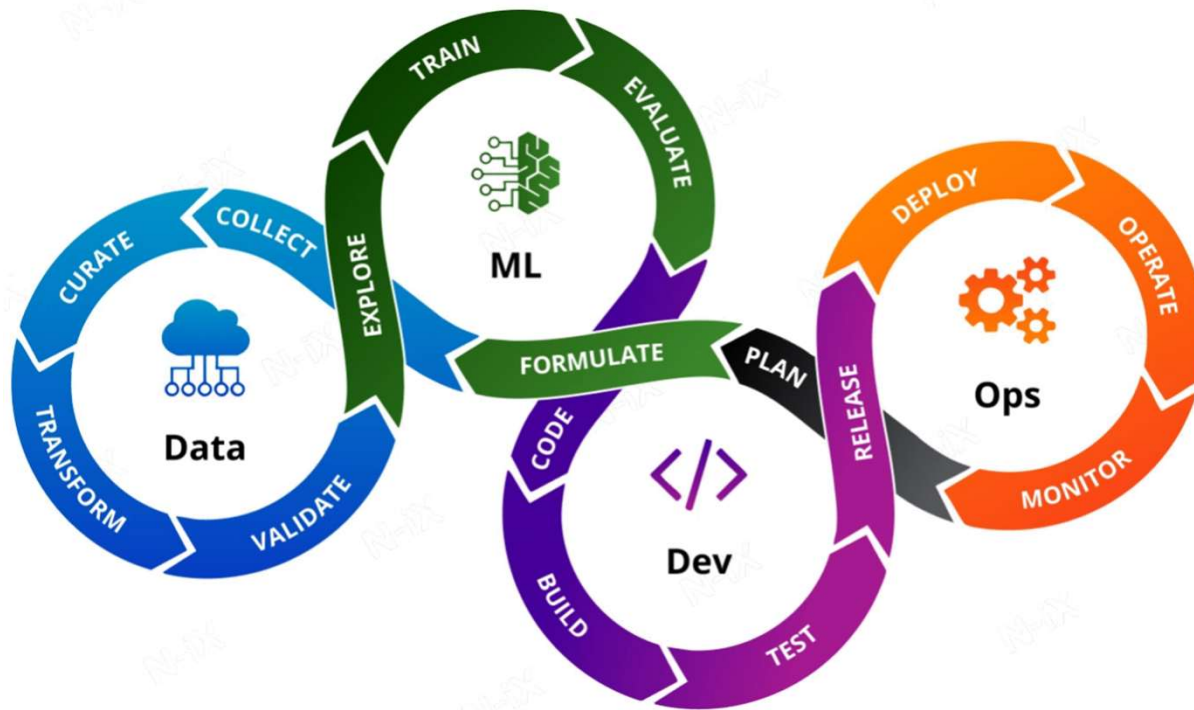
Machine Learning Operations (MLOps)

Productionizing Models



Source: martinowler.com

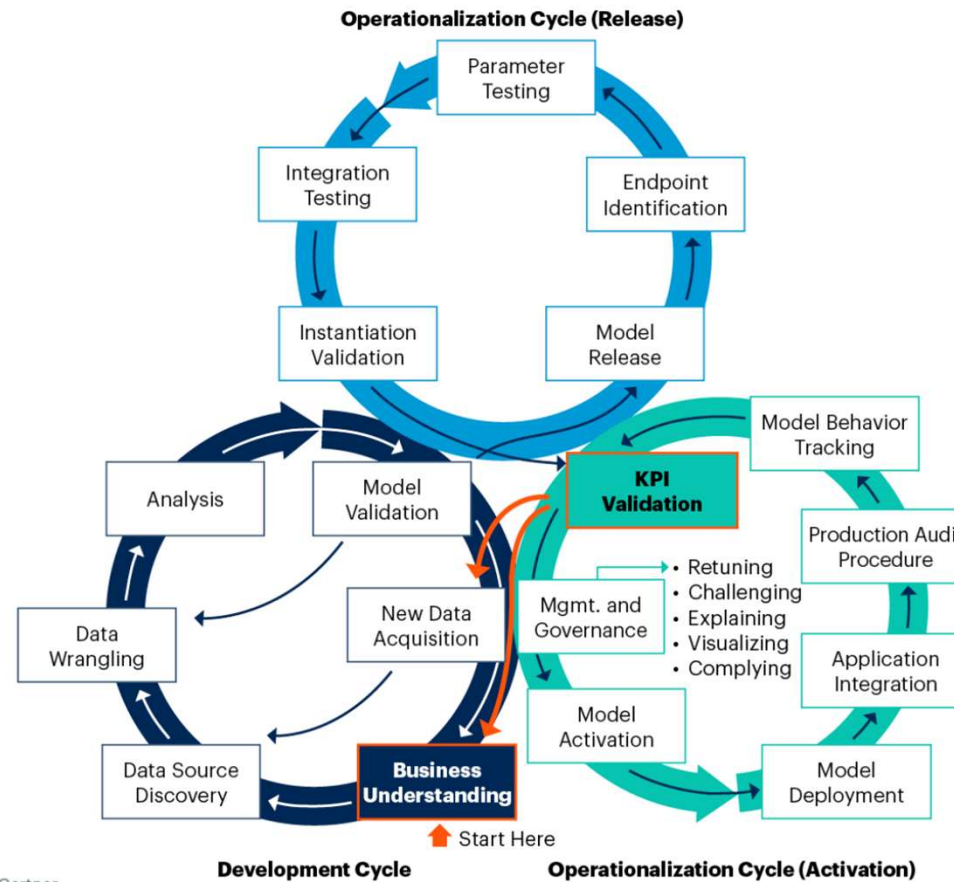
MLOps & DevOps



Source: n-ix

MLOps & DevOps

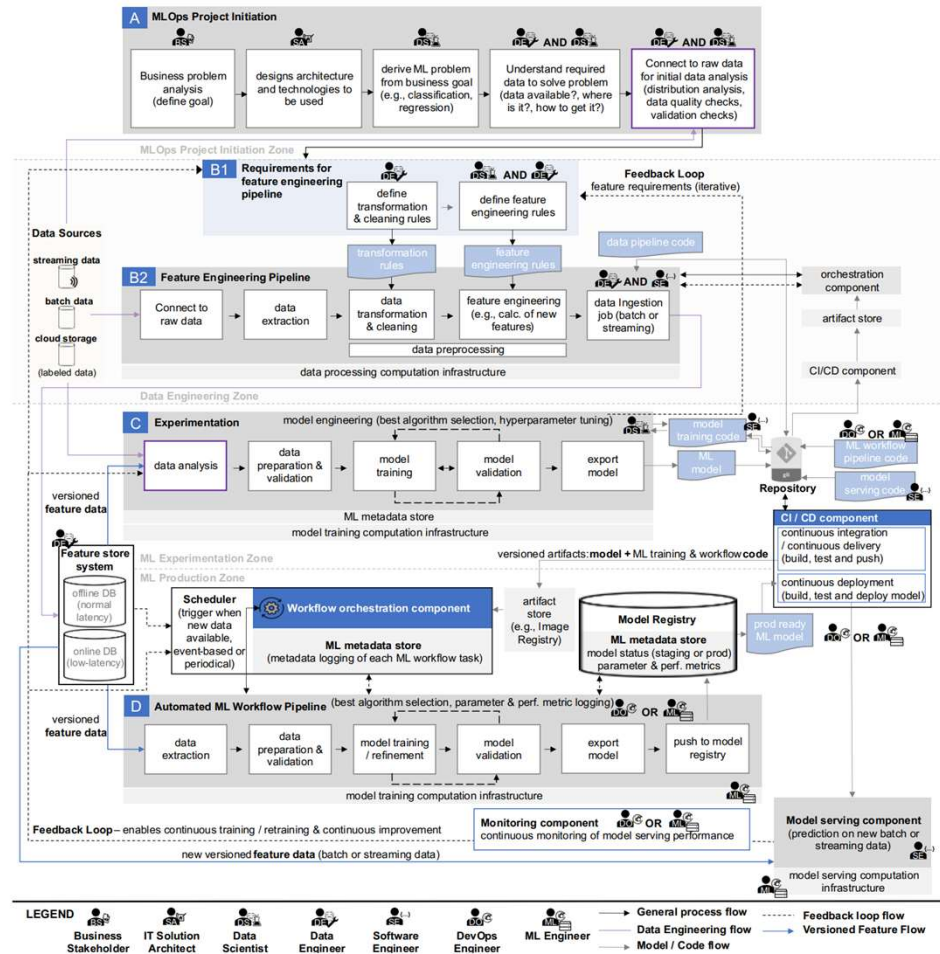
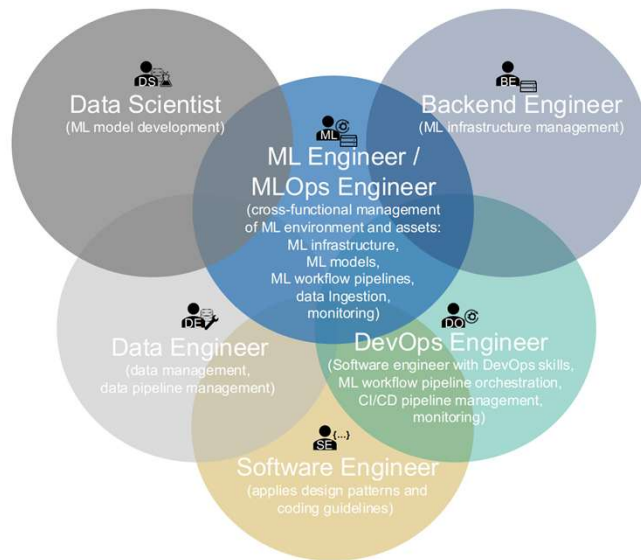
Gartner's MLOps Framework



Source: n-ix

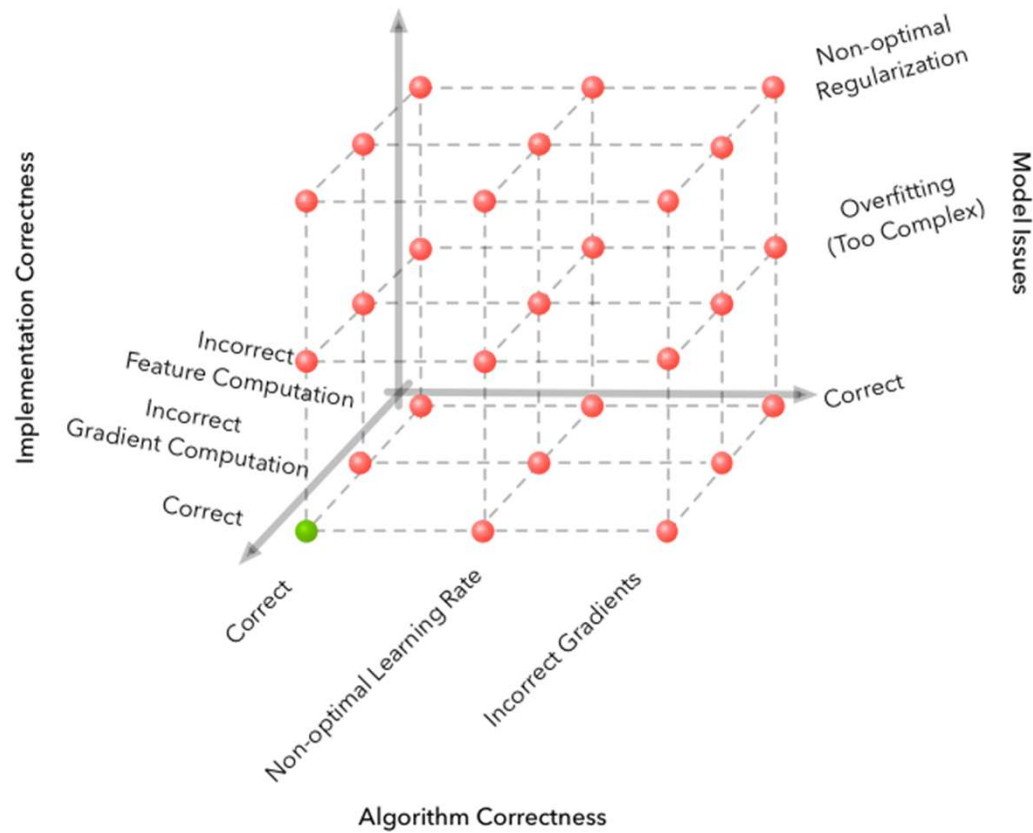
Source: Gartner

MLOps & DevOps



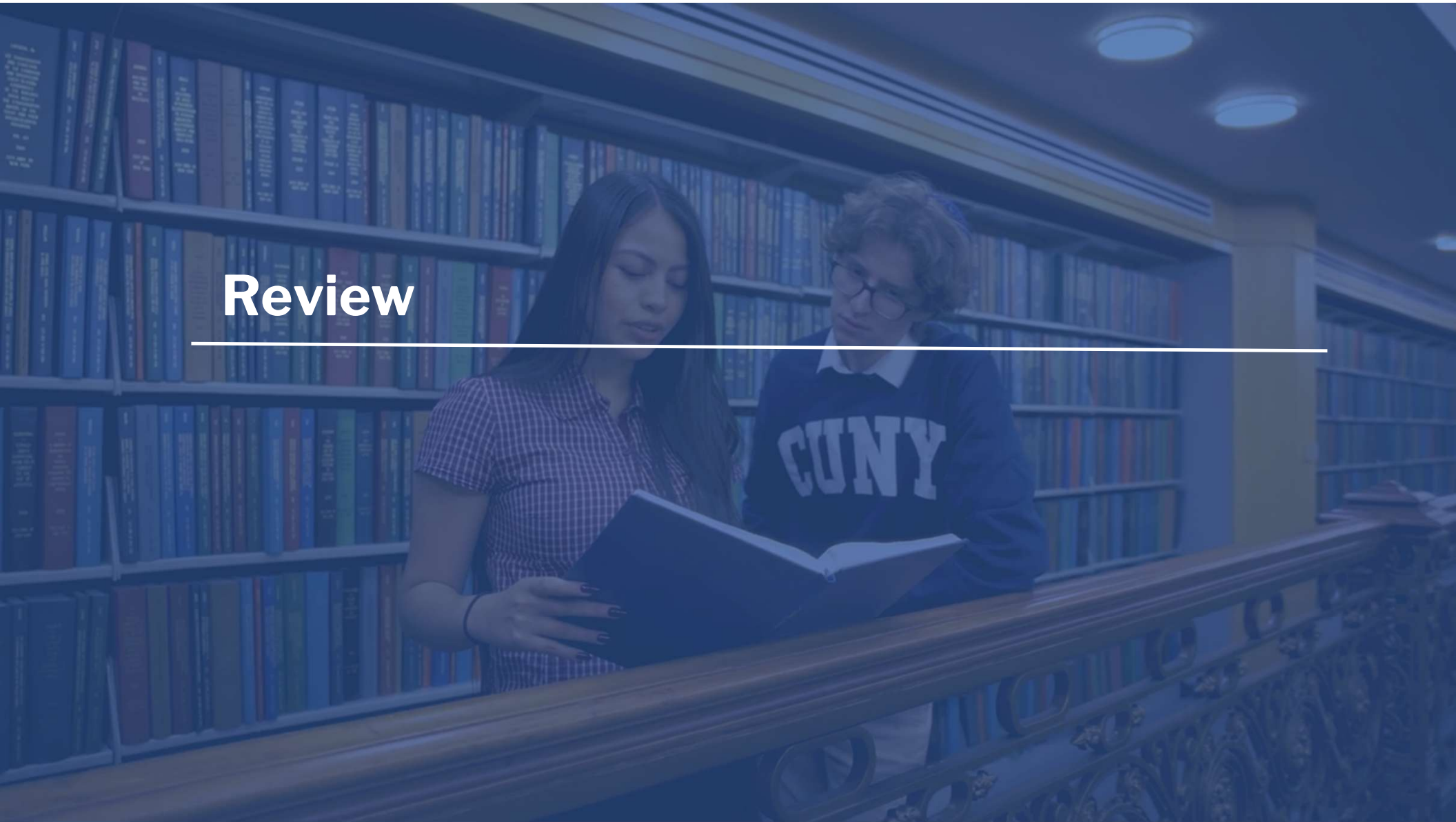
Source: "Machine Learning Operations (MLOps): Overview, Definition, and Architecture", Kreuzberger, et al

MLOps & DevOps



Source: n-ix

Review



This week we covered

Lesson Objectives/Topics

- **Foundational**

- Implement ensemble methods like bagging and random forests for classification tasks.
- Analyze the impact of aggregation on prediction accuracy and stability.
- Evaluate ensemble model performance using business-oriented KPIs.
- Explain the advantages of ensemble learning in high-variance environments.

- **Implementation & Strategy**

- Machine Learning Development Lifecycle (ML-SDLC) & MLOps

An aerial photograph of the Manhattan skyline, featuring the Manhattan Bridge in the foreground and the dense cityscape in the background. The bridge's stone towers and suspension cables are prominent. The water of the Hudson River is visible in the foreground. The sky is clear and blue.

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